

QSpikeXAI-Net

A Quantum-Enhanced Spiking Neural Network with Multi-Level Explainability for Unified EEG-Based Neurological Disorder Classification

Research Architecture & Execution Plan

Target: Q1 Journal (IEEE TNSRE / NeuroImage / Neural Networks)

Last Updated: May 2026 | Platform: A100 GPU Server

1. Executive Summary & Core Novelty

This document defines the complete research plan for a Q1 journal submission. The project extends the existing EEG codebase (which currently achieves up to 94.7% accuracy on 5-class sleep staging with SNN-1D-Attention) into a unified, multi-disorder framework covering Sleep Apnea, Schizophrenia, and Mild Cognitive Impairment (MCI). The proposed architecture introduces three tightly coupled novelties that no prior published work has combined:

Three Core Novelty Pillars

- Quantum-SNN Dual Stream:** Parallel raw-EEG spiking (SNN-1D-Attn) and spectrogram-based quantum (VQC-ViT) streams, fused via a confidence-weighted conditional router.
- Quantum-Level XAI:** Explainability extended into the quantum circuit itself — qubit-wise attribution scores show WHICH quantum parameters drove a classification decision.
- Unified Multi-Disorder Framework:** A single shared backbone with disorder-specific heads, trained and evaluated simultaneously across Sleep Apnea, Schizophrenia, and MCI.

2. Problem Statement

EEG-based classification of neurological and psychiatric disorders is individually studied but rarely unified. Each disorder has distinct EEG biomarkers:

Disorder	Key EEG Biomarkers & Challenges
Sleep Apnea	Respiratory-linked EEG arousal patterns, oxygen desaturation correlates; intermittent artifacts from movement. 4-class severity: Healthy, Mild, Moderate, Severe.
Schizophrenia	Reduced gamma-band power (30-80 Hz), disrupted alpha coherence, altered P300 ERP; high inter-subject variability; binary or 3-class (SZ / FEP / HC).

Mild Cognitive Impairment	Increased theta/delta ratio, slowing of dominant frequency, reduced alpha peak; progression risk marker; binary (MCI vs. HC) or 3-class (HC / MCI / AD).
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A unified model must handle heterogeneous class structures, different recording paradigms (resting-state vs. event-related), variable channel counts, and diverse clinical benchmarks. This multi-task framing is the entry point for a high-impact Q1 contribution.

3. Datasets

All datasets listed below are publicly accessible with no special data-use agreement. Download commands and accession IDs are provided for each.

TASK 1: Sleep Apnea

Dataset	Details & Download
PhysioNet Apnea-ECG (ALREADY DOWNLOADED)	70 overnight recordings (ECG + apnea annotations). Binary apnea/no-apnea per minute + per-recording AHI severity. Path: data/apnea-ecg-database-1.0.0/ URL: physionet.org/content/apnea-ecg/1.0.0/
PhysioNet SHHS (Sleep Heart Health Study)	5804 overnight PSG recordings. Gold-standard for sleep apnea severity. EEG channels: C3-A2, C4-A1 + oximetry. Download: physionet.org/content/shhs/1.0.0/ Credential-free subset via: <code>wget -r https://physionet.org/files/shhs/1.0.0/</code>
BOAS / OpenNeuro ds005555 (REFERENCED IN PROJECT)	Sleep staging dataset already used by pipeline.py. Provides complementary 5-class sleep stages. BIDS format. Download: openneuro.org/datasets/ds005555

TASK 2: Schizophrenia

Dataset	Details & Download
PhysioNet EEG Schizophrenia (Olejarczyk & Jernajczyk 2017)	14 schizophrenia patients + 14 healthy controls. 19-channel EEG, 250 Hz, eyes-closed resting state, ~15 min per subject. Canonical benchmark — used in 50+ papers. Download: physionet.org/content/eeg-schizophrenia/1.0.0/ <code>wget -r https://physionet.org/files/eeg-schizophrenia/1.0.0/</code>
Phinyomark Schizophrenia EEG (OpenNeuro ds003774)	84 subjects (42 SCZ, 42 HC). 32 channels, 1000 Hz. Eyes-open/closed conditions. Larger and more recent benchmark. Download: openneuro.org/datasets/ds003774 <code>openneuro download ds003774</code>

COBRE EEG (SchizConnect)	~70 subjects with first-episode psychosis, chronic SZ, and HC. Resting-state EEG + MRI. Available via SchizConnect portal. URL: schizconnect.org (free registration) Primarily for cross-modal validation.
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Recommended starting point: PhysioNet ds (14+14 subjects) for fast prototyping, then scale to ds003774 for final evaluation. Use binary labels (SCZ vs HC) initially, optionally extend to First-Episode Psychosis (FEP) as a 3-class variant.

TASK 3: Mild Cognitive Impairment

Dataset	Details & Download
CAUEEG (Korean MCI Dataset) — Primary	55 HC, 55 MCI subjects. 19-channel resting EEG, 512 Hz. BIDS format. Publicly available, no registration. Paper: Kim et al. 2022, Scientific Data. Download: github.com/ipis-mjkim/caueeg-dataset
OpenNeuro ds002778 (MCI Auditory Oddball)	50+ subjects (HC, MCI, AD). P300 event-related paradigm — strong cognitive biomarker. 64 channels, 1000 Hz. Download: openneuro.org/datasets/ds002778 openneuro download ds002778
PhysioNet EEG During Mental Arithmetic (Baseline / transfer)	36 subjects performing mental arithmetic. Cognitive load paradigm. Useful as a transfer/fine-tuning set. URL: physionet.org/content/eegmat/1.0.0/

The CAUEEG dataset is the recommended primary benchmark for MCI because it is the largest freely available resting-state EEG MCI dataset with BIDS formatting and an established prior-work baseline, making comparison straightforward.

4. Architecture Framework: Three Tiers

The experimental design follows an escalating complexity ladder. Each tier is necessary — it provides ablation evidence for the paper and demonstrates progressive improvement.

4.1 Tier 1 — Baselines

These are standard, well-cited models that anchor the paper's comparison table. All already exist or are trivially re-usable from the existing codebase.

BASELINE	Tier 1: Standard Deep Learning Baselines
EEGNet	Compact depthwise + separable CNN. The canonical EEG classification baseline (Lawhern et al. 2018). 3,274 params. Already exists in literature; cite SHU paper benchmarks.

EEG-TCNet	EEGNet + Temporal Convolutional Network (TCN). Strong temporal baseline for time-series. Low complexity.
1D-CNN / ResNet18	Raw-signal 1D ResNet. Already in project (sleep_apnea_pipeline.py). Transfer from ImageNet weights via 1D adaptation.
ViT-1D Baseline	Standard Vision Transformer applied to 1D EEG segments. Already in project as snn_vit. Serves as non-spiking transformer anchor.
SNN-LIF-ResNet (2D)	Spiking ResNet on CWT scalogram. Already in project. Best local JSON: swin ~84.6%.

4.2 Tier 2 — Mid-Level Ablation Models

These are enhanced variants that isolate individual contributions. They form the ablation table and bridge Tier 1 and Tier 3.

MID-LEVEL	Tier 2: Enhanced / Ablation Models
SNN-1D-Attn	Raw EEG SNN with LIF neurons + temporal attention. Best W&B result: 94.7% acc, 0.84 macro-F1. This is the STRONGEST SINGLE-STREAM model in the project. Use as the SNN-only ablation point.
Quantum-1D-Ring-RYZ	1D quantum hybrid with ring entanglement + RYZ encoding. Best quantum W&B result: 94.6% acc. Use as the quantum-only ablation.
EEG-MFTNet (Multi-scale + Transformer)	Architecture from reference paper. Multi-scale temporal convolutions (kernels $k=\{5,9,13,29,61,125\}$) + Transformer encoder. Achieved 58.9% on SHU MI task. Adapt to resting-state for our tasks.
Dual-Stream Fusion (SNN+Quantum, no gate)	Concatenate SNN-1D-Attn and Quantum-1D features without conditional routing. Fixed-weight fusion. Ablation baseline for the router.
SNN+Quantum+XAI (no quantum attribution)	Full fusion with standard Integrated Gradients XAI on inputs/channels. No quantum-level attribution. Shows value of adding quantum attribution layer in Tier 3.

4.3 Tier 3 — Proposed Novel: QSpikeXAI-Net

This is the main contribution. It assembles the strongest components from the project into a coherent novel architecture, adds the conditional routing gate, and extends XAI into the quantum layer.

QSpikeXAI-Net: Full Architecture Specification

Component A — Dual-Stream Input Processing

Stream	Processing Pipeline
Stream 1: Temporal (SNN path)	Raw EEG segments (channels $\times$ T) $\rightarrow$ Band-pass filter (0.5–45 Hz) $\rightarrow$ Z-score normalization per channel $\rightarrow$ LIF-SNN encoder with temporal attention (from src/models/snn_1d/) $\rightarrow$ Output: temporal feature vector $F_{\text{snn}} \in \mathbb{R}^{256}$

Stream 2: Spectral (Quantum path)	Raw EEG → Continuous Wavelet Transform (Morlet, 40 scales) → CWT scalogram (channels × freq × time) → Patch embedding (ViT-style, patch_size=8) → Variational Quantum Circuit (VQC) layer (4 qubits, 3 rotation layers) → Output: quantum feature vector $F_{qnn} \in \mathbb{R}^{128}$
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Component B — Variational Quantum Circuit (VQC) Layer

Implemented using PennyLane (simulation mode on A100) or a pure-PyTorch custom VQC as in the reference quantum_transformer_ViT_cifar10.py. The VQC replaces the standard MLP bottleneck in the ViT-patch path:

Parameter	Value / Description
n_qubits	4 (16-dimensional Hilbert space per head)
n_heads	4 (parallel VQC heads, each processes a feature subset)
Encoding	Angle encoding: classical features → RY rotation angles
Entanglement	Ring topology (CNOT: 0→1, 1→2, 2→3, 3→0). Matches best quantum result in project (quantum_ring_RYZ: 94.56%)
Rotation layers	3 layers of [RX, RY, RZ] rotations per qubit (learnable via PyTorch Parameter)
Measurement	Z-basis expectation values → classical output vector
Parameters	~576 learnable (3 layers × 4 qubits × 3 gates × 4 heads + linear layers)

Note: Use the VQCLayer implementation pattern from the reference quantum_transformer_ViT_cifar10.py (arch-eval-088), adapted from image patches to EEG scalogram patches. The WaveletKANLayer (KAN FFN) from the same file can replace the standard MLP in the fusion stage — this is an additional novelty point.

Component C — Confidence-Gated Conditional Router

Replaces simple concatenation. Learns WHEN to trust the SNN path vs. the quantum path:

Element	Description
Gate input	Concatenate $[F_{snn} F_{qnn}] \rightarrow \text{Linear}(384, 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, 2) \rightarrow \text{Softmax}$
Gate output	$\alpha = [\alpha_{snn}, \alpha_{qnn}]$, summing to 1.0
Fusion	$F_{fused} = \alpha_{snn} \times W_{snn}(F_{snn}) + \alpha_{qnn} \times W_{qnn}(F_{qnn})$ where W_{snn}, W_{qnn} are linear projections to common dim 256

Conditional routing	At inference, α can be logged: high α_{qnn} = spectral patterns dominant; high α_{snn} = temporal dynamics dominant. This is itself an interpretable biomarker.
From project	Directly extends conditional_routing_fusion.py from src/models/fusion/ — already proven design.

Component D — Task-Specific Classification Heads

Task	Head Architecture & Labels
Sleep Apnea	F_fused $\rightarrow$ Linear(256,128) $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(128, 4) $\rightarrow$ Softmax Classes: Healthy Mild Moderate Severe (AHI-based)
Schizophrenia	F_fused $\rightarrow$ Linear(256,64) $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(64, 2) $\rightarrow$ Softmax Classes: Healthy Control Schizophrenia (extend to 3-class with FEP if ds003774)
MCI	F_fused $\rightarrow$ Linear(256,64) $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(64, 3) $\rightarrow$ Softmax Classes: HC MCI AD (or binary HC MCI for ds002778 protocol)

Component E — Multi-Level XAI Module

XAI is not an afterthought. It is computed as part of the forward pass and is a separate evaluation section in the paper. Four attribution levels:

XAI Level	Method & Implementation
L1: Channel Attribution	Integrated Gradients on input EEG channels. From signal_xai.py in QuantumNeuroXAI repo: integrated_gradients_signal(model, x, baseline=zeros, steps=32). Output: per-channel importance score.
L2: Temporal Attribution	Transformer attention rollout over the temporal tokens in the ViT-patch stream. Visualize as a time-series heatmap overlaid on the scalogram.
L3: Frequency Band Attribution	channel_band_summary() from QuantumNeuroXAI: decompose saliency map into delta/theta/alpha/beta/gamma band contribution scores. Maps directly to known neurophysiological biomarkers.
L4: Quantum Gate Attribution	Novel contribution. Compute gradient of output w.r.t. each VQC rotation parameter $\theta_{\{\text{layer}, \text{qubit}, \text{gate}\}}$. Produces a 3D attribution tensor [layers $\times$ qubits $\times$ gates]. Visualize as a heatmap: which rotation gates (RX/RY/RZ) and which qubits drove the decision. This is the quantum explainability contribution.

The quantum attribution is straightforward with autograd since all VQC operations are differentiable PyTorch ops. The paper section on XAI demonstrates that the quantum circuit attends to neurophysiologically meaningful frequency features, validating the model's trustworthiness for clinical use.

5. Implementation Plan

5.1 Dataset Loaders to Build

Three new dataset loaders are needed, following the pattern of `boas_dataset.py` and `apnea_ecg_dataset.py`:

File to Create	What It Does
<code>src/data/schizophrenia_dataset.py</code>	Loads PhysioNet EEG-schizophrenia EDF files. Outputs 10-sec epochs (2500 samples @ 250 Hz). Applies z-score + bandpass 0.5–45 Hz. Subject-level 70/15/15 split.
<code>src/data/caueeg_dataset.py</code>	Loads CAUEEG dataset (BIDS format). 19 channels, 512 Hz. Epochs of 8 sec (4096 samples). Resamples to common 256 Hz. Labels: HC=0, MCI=1.
<code>src/data/mci_erp_dataset.py</code>	Loads ds002778 P300 ERP data. Segments around stimulus onset (-200ms to 800ms). Uses P300 amplitude at Pz as feature engineering option.
<code>src/data/unified_loader.py</code>	Wrapper that instantiates the correct dataset class based on task flag and returns standardized (epochs, labels) tensors with a common channel normalization to 19 channels.

5.2 Model Code Structure

File	New Code / Modification
<code>src/models/quantum/vqc_layer.py</code>	Standalone VQC layer. Adapt from <code>quantum_transformer_ViT_cifar10.py</code> : <code>VQCLayer</code> class with <code>n_qubits=4</code> , 3 rotation layers, ring CNOT entanglement. Pure PyTorch (no PennyLane dependency for speed).
<code>src/models/quantum/wavelet_kan.py</code>	<code>WaveletKANLayer</code> and <code>WaveletFFN</code> from reference file. Optional FFN replacement.
<code>src/models/qspikexai_net.py</code>	Main <code>QSpikeXAI-Net</code> model. Combines SNN-1D-Attn stream, VQC-patch stream, conditional router, task heads. Inherits from <code>nn.Module</code> .
<code>src/models/xai/quantum_xai.py</code>	<code>quantum_gate_attribution(model, x)</code> : compute gradient of output w.r.t. VQC theta parameters. Returns attribution tensor [3, <code>n_qubits</code> , 3].

src/models/xai/signal_xai.py	Copy from QuantumNeuroXAI repo. Provides input_saliency, integrated_gradients_signal, channel_band_summary.
unified_pipeline.py (extend)	Add loaders and model entries for schizophrenia + MCI tasks. This is the right file to evolve — do NOT add more logic to pipeline.py.

5.3 Training Configuration

Parameter	Value
Optimizer	AdamW, lr=1e-3, weight_decay=1e-4
LR Schedule	CosineAnnealingLR (T_max=50 epochs) with warmup (5 epochs linear)
Loss	Cross-entropy with label smoothing=0.1 (from reference paper)
Batch size	128 (A100 can handle easily)
Epochs	50–100 (use checkpoint on best val macro-F1 not accuracy)
Class weights	Compute from label distribution — especially critical for MCI (imbalanced)
Data augmentation	EEG-specific: random temporal crop, Gaussian noise ($\sigma=0.05$), channel dropout ($p=0.1$), random frequency masking
Cross-validation	5-fold subject-level CV (not trial-level — prevents data leakage)
Mixed precision	fp16 via torch.cuda.amp (already in ResearchConfig)
Gradient clipping	max_norm=1.0 (from reference paper)

6. Experiments & Expected Paper Tables

6.1 Main Result Table (Comparison with Baselines)

This is the centerpiece table of the paper. Format: accuracy $\pm$ std, macro-F1 $\pm$ std, AUC across all three tasks.

Model	SA Acc%	SZ Acc%	MCI Acc%	Avg F1	Params
EEGNet	~72	~78	~74	~0.73	3.3K
EEG-TCNet	~76	~82	~77	~0.77	4.1K
1D-ResNet	~79	~85	~80	~0.80	11M
VIT-1D	~81	~87	~82	~0.82	5.2M

SNN-1D-Attn (Tier 2)	~87	~91	~88	~0.88	2.1M
Quantum-1D-Ring (Tier 2)	~86	~90	~87	~0.87	1.8M
EEG-MFTNet (ref. paper)	~83	~88	~84	~0.84	16K
Dual-Stream (no gate)	~89	~93	~90	~0.90	3.9M
QSpiceXAI-Net (proposed)	~93	~96	~93	~0.94	4.2M

SA = Sleep Apnea, SZ = Schizophrenia, MCI = Mild Cognitive Impairment. Numbers are expected ranges based on literature and existing project results. Actual numbers will populate from experiments. Note: SNN-1D-Attn already achieves 94.7% on sleep staging (5-class), so these estimates for binary/4-class tasks are conservative.

6.2 Ablation Table

Configuration	What It Tests
SNN stream only (no quantum)	Value of quantum stream
Quantum stream only (no SNN)	Value of temporal SNN stream
Both streams, no gate (concat)	Value of conditional router
Both streams + gate, no XAI loss	XAI-awareness during training (if regularization used)
Full QSpiceXAI-Net	Complete model — all components contribute

6.3 XAI Evaluation

Evaluation	Method
Channel importance faithfulness	Electrode deletion test: progressively zero out top-K and bottom-K channels, measure confidence drop. Expect clear separation (as in reference paper Fig. 4).
Frequency band alignment	Check that delta/theta bands dominate for sleep apnea; gamma bands for schizophrenia; alpha-slowness for MCI — consistent with neurophysiology literature.
Quantum gate attribution	Visualize VQC θ attribution heatmap. Show cross-disorder differences in which rotation gates activate.
Router gate analysis	Plot gate weights α_{snn} vs α_{qnn} distribution per disorder. Hypothesis: SZ (short event patterns) $\rightarrow$ higher α_{snn} ; Sleep apnea (long spectral patterns) $\rightarrow$ higher α_{qnn} .

7. Immediate Codebase Cleanup (Before Expansion)

The project notes itself that several things must be resolved before adding more parallel paths. These are the concrete actions to take in the first week:

Issue	Action
pipeline.py and unified_pipeline.py diverged	DO NOT add any new models or tasks to pipeline.py. All new work goes into unified_pipeline.py. Gradually backport the full model registry from pipeline.py into unified_pipeline.py.
Results split across 4 locations	Create a single canonical_results.csv with columns: task, model, dataset, fold, accuracy, macro_f1, balanced_acc, auc, epoch, command, git_hash, date. Merge allResults.csv and results/*.csv into it immediately.
BOAS dataset missing locally	Either download it (openneuro.org/datasets/ds005555) or update DATASETS.md to flag it as needed. Do not run pipeline.py experiments that require it until confirmed.
error.json in quantum_ring and snn_lif_vit folders	Before relying on those result.json accuracies, rerun with --epochs 1 --max-subjects 5 to confirm clean execution. The num_samples=0 error suggests a split/cache state issue.
tcanet import path mismatch	Fix import in pipeline.py: change 'from tcanet_clean import' to 'from src.models.tcanet_clean import'. Verify with: <code>python -c 'from src.models.tcanet_clean import TCANetClean; print(ok)'</code>
Apnea code duplication	Keep sleep_apnea/ package as the canonical implementation. sleep_apnea_pipeline.py can be a thin CLI wrapper that calls sleep_apnea/ internals.

8. Execution Timeline

Week	Deliverable
Week 1	Codebase cleanup (Section 7). Download all three datasets. Write unified_loader.py. Verify all existing experiment runs in canonical_results.csv.
Week 2	Build schizophrenia_dataset.py and caueeg_dataset.py. Run all Tier 1 baselines on all 3 tasks. Populate comparison table rows 1–5.
Week 3	Implement vqc_layer.py (port from reference file). Build QSpikexAI-Net skeleton (qspikexai_net.py) with dummy heads. Verify forward pass on all 3 tasks.

Week 4	Train Tier 2 models (SNN-1D-Attn, Quantum-1D, EEG-MFTNet) on all 3 tasks. Populate rows 6–8 of comparison table.
Week 5	Full QSpikexAI-Net training run on all tasks (5-fold CV). Collect results. Begin XAI computation.
Week 6	XAI visualizations: channel deletion tests, frequency band summaries, quantum gate attribution heatmaps. Write XAI evaluation section.
Week 7–8	Paper writing. Introduction, Related Work, Methods (with architecture diagram), Results, Discussion.
Week 9	Internal review, revision, proofreading. Submit to target journal.

9. Target Journals (Q1)

Journal	Fit & Notes
IEEE TNSRE (Trans. Neural Systems & Rehab Eng.)	Perfect fit: EEG, BCI, neurological disorders, deep learning. IF ~4.5. Fast review. STRONGEST recommendation for this work.
Neural Networks (Elsevier)	High IF (~6), covers quantum ML, SNNs, XAI. Broader scope suits the novel quantum+SNN combination.
Computers in Biology and Medicine	IF ~7, clinical AI focus. Good for the multi-disorder unified framing and XAI for clinical interpretability.
NeuroImage	Prestige journal for neuroimaging + methods. IF ~5.7. Suitable if XAI visualization quality is strong.
Journal of Neural Engineering (IOP)	High IF, strong track record for EEG classification papers. Good alternative to TNSRE.

10. Quick Decision Map

When to touch which file
New dataset? → <code>src/data/<disorder>_dataset.py</code> + <code>src/data/unified_loader.py</code>
New model component? → <code>src/models/qspikexai_net.py</code> or <code>src/models/quantum/vqc_layer.py</code>
New XAI method? → <code>src/models/xai/</code>

New experiment run? → unified_pipeline.py (NOT pipeline.py)

New result? → append to canonical_results.csv immediately

Paper table update? → canonical_results.csv is the single source of truth

Architecture diagram? → draw QSpiceXAI-Net with two parallel streams, router gate, three task heads, four XAI layers